

256-Note Sequencer

This digital sequencer for voltage-controlled music synthesizers can store and automatically play back up to 256 notes.

In principle, the sequence of voltages generated when playing the keyboard is digitized by an 8-bit A/D converter and then stored. Additionally, note durations and the lengths of pauses between notes (if any) are converted into proportional time signals using a clock generator and two counters, and are also stored. During playback of the stored note sequence(s), a D/A converter transforms the 8-bit word—which corresponds to the keyboard voltage—back into the original voltage. The "encoded" note and pause durations are processed by counters that convert them back to their original state. This method of signal processing offers advantages over other systems that use an "encoded keyboard" for the following reasons:

- 1. The existing keyboard does not need to be modified.
- 2. An 8-bit A/D converter is less expensive than an "encoded keyboard."
- 3. The use of a dual-mode converter (A/D and D/A) ensures high playback accuracy.

Furthermore, the circuit presented here requires less memory than others that enter keyboard information into storage at a fixed clock frequency, as it evaluates only the durations of notes and pauses. The circuit operates as follows: To record a sequence of notes, switch S1 to the "Record" position, which simultaneously sets the memory to the "Write" position. The system is now ready to record. The recording process begins when a key on the keyboard is pressed. The resulting gate pulse (indicating that a key has been pressed) sets flip-flop N11/N12.

Consequently, a logic 0 is applied to one of the data inputs of IC18, which simultaneously releases the reset state of IC19. The address counter can then begin cycling through the memory addresses. The gate pulse also triggers a "dynamic shift register" (constructed from N20–N24) that performs the following functions: resetting counters IC10 ("note counter") and IC13 ("pause counter"), resetting the ZN425 A/D-D/A converter, clearing memory IC18, and clocking address counter IC19. The result of these commands is that the first memory location remains empty.

The keyboard voltage is digitized by IC ZN425. IC10 counts the clock pulses from IC9 based on the note duration. When the key is released, IC10 similarly counts the clock pulses from IC9 based on the duration of the rest. Thus, when a key is pressed again, the storage process begins, and the previously obtained information (note or rest) is placed in the second memory location. Of course, the note duration of the gate pulse is not always precisely defined, especially when playing legato. For this reason, a detector (IC4, IC5) is included to sense changes in the keyboard voltage. It consists of a differentiating network followed by a window comparator that generates a negative pulse whenever the keyboard voltage changes. When either IC10 or IC13 reaches its maximum counting capacity (6 bits), the output of N4 or N9 goes to logic 0; this initiates a new storage process and resets the counters. In this way, a single note or rest can span two or more memory locations.

It was previously stated that the first memory location remains empty. However, if a pause is desired at the beginning of the melody, this memory location can also be used. To do this, S3 is pressed, activating ICs 10, 13, and 18 independently of the keyboard's gate pulse. IC 13 then counts the pulses from the clock generator until a key is pressed. At that point, the previously described reset and set operations take place, and the 6-bit word from the counter—representing the pause—is loaded into the first memory location of ICs 12 and 15.

In playback mode, switch S1 is in the 'Play' position, the memory is in the 'Read' position, and the ZN425 converter is in D/A mode. To retrieve the note and/or rest durations, IC10 is clocked until its output data matches the data output from IC12 and IC15.

During this time, the gate pulse output is at a logic high level (logic 1). Upon data matching, the gate pulse output switches to a logic low level (logic 0). Clock pulses are then routed to IC13. When the data at its output matches the data outputs of IC15 and IC17, IC19 initiates the retrieval of the next memory addresses. The stored data is sent back to the keyboard via the D/A converter. Playback of the stored melody ends when the output of IC18 (pin 12) goes high (logic 1). IC19 is then reset, unless S2 is set to the "Repeat" position. In that case, IC19 resets upon the arrival of the logic high signal, the address counter begins accessing the lowest address, and the melody starts over. The unused I/O pins of IC18 can be utilized for control functions. Control signals for various VCOs, filters, etc., are connected here (via appropriate buffers, of course).

Two additional notes:

- 1. Portamento controls and voltage controls (coarse and fine) should be connected after the sequencer.
- 2. The A/D converter can process data from a 3-octave keyboard. A 4-octave keyboard can be used if the feedback resistor of op-amp IC13 is increased from 10 k Ω to 18 k Ω .

IC6 is a Ferranti IC, available through Ferranti distributors.

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Literature: T. Emmens, Elektor Semiconductor Issue 1979, pp. 8–28, Elektor Verlag





